

Supplementary Material

Teacher-Health-Guided Speech Release

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I. THEORETICAL DETAILS

A. Privacy of the Complete Adaptive Release

For the complete release map in main III-A, let $K(z, A) \in [0, 1]$ be its conditional release probability given the accounted teacher output z . Integrating the DP inequality gives $\int K dP_D \leq e^\epsilon \int K dP_{D'} + \delta$ [1]. This includes adaptive checkpoint/route selection and student training, conditional on permitted side information. Adaptively choosing Hard retains selection's private dependence; all inspected private training must be accounted.

B. Balanced Softmax: A Pre-clipping Identity

Differentiation gives $(p_y - 1)h$ for CE and $(q_y - 1)h$ for BS [2], where $q_y = \nu_y p_y / \sum_j \nu_j p_j$. Thus $\nu_y < \sum_j \nu_j p_j$ increases the target-row norm for $h \neq 0$, before clipping; both norms vanish at $h = 0$. Training-time logit adjustment with $\tau = 1$ equals BS [3].

For CV17's linear head including bias, the complete gradient is the outer product $(q - e_y)[h; 1]^\top$. Any probability vector satisfies $\|q - e_y\|_2 \leq \sqrt{2}$. Unit-normalized features therefore give Frobenius norm at most $\sqrt{2}\sqrt{1 + \|h\|_2^2} = 2$, explaining the chosen clipping norm. The same bound holds for CE with p in place of q . It is specific to the normalized linear head, not the earlier mel-CNNs; those models' recorded clipping fractions can be near one without contradicting this bound.

C. Health Estimation and Region Probabilities

For a fixed teacher and independent calibration draws, let $\theta_{ij} = \Pr(Y = i, \hat{Y}_t = j)$ and $\hat{\theta} = C/n$. Write $r_k = \sum_j \theta_{kj}$, $c_k = \sum_i \theta_{ik}$, and $d_k = \theta_{kk}$. Assume $r_k > 0$ and a unique maximum column k^* when $\lambda_3 > 0$. The delta method gives

$$\hat{\sigma}_H^2 = \frac{1}{n} \nabla H(\hat{\theta})^\top [\text{Diag}(\hat{\theta}) - \hat{\theta} \hat{\theta}^\top] \nabla H(\hat{\theta}). \quad (14)$$

The derivatives are

$$\frac{\partial B}{\partial \theta_{ab}} = \frac{\mathbf{1}\{a = b\} r_a - d_a}{K r_a^2}, \quad (15)$$

$$\frac{\partial F}{\partial \theta_{ab}} = \frac{1}{K} \sum_k \left[\frac{2\mathbf{1}\{a = b = k\}}{r_k + c_k} - \frac{2d_k(\mathbf{1}\{a = k\} + \mathbf{1}\{b = k\})}{(r_k + c_k)^2} \right], \quad (16)$$

$$\frac{\partial(1 - M)}{\partial \theta_{ab}} = -\mathbf{1}\{b = k^*\}. \quad (17)$$

Combine them as $\nabla H = \lambda_1 \nabla B + \lambda_2 \nabla F + \lambda_3 \nabla(1 - M)$. Macro-F1's expression $2d_k/(r_k + c_k)$ avoids a false singularity at zero true positives. Tied maxima and same-development selection fall outside this regular fixed-teacher approximation.

Proposition 2 (Confidence margins and routing stability). *For the Gaussian working model, $\hat{\sigma}_H > 0$, and normal quantile z_p , the exact KD and Hard region certificates are*

$$\begin{aligned} \hat{H} - z_{1-\rho} \hat{\sigma}_H &\geq \tau_{\text{high}} & (\text{KD}), \\ \hat{H} + z_{1-\rho} \hat{\sigma}_H &\leq \tau_{\text{low}} & (\text{Hard}). \end{aligned} \quad (18)$$

HAKD requires its full interval mass to be at least $1 - \rho$. Off both thresholds, let $d = \min\{|\hat{H} - \tau_{\text{low}}|, |\hat{H} - \tau_{\text{high}}|\}$. Then

$$1 - q_{m_0(\hat{H})} \leq \min \left\{ 1, 2 \exp \left(-\frac{d^2}{2\hat{\sigma}_H^2} \right) \right\}, \quad (19)$$

so $q_{m_0(\hat{H})} \rightarrow 1$ as $\hat{\sigma}_H \rightarrow 0$.

Inverting the Gaussian CDF gives the margins. Leaving the point estimate's region requires a deviation of at least d ; the two-sided normal tail bound proves stability. The masses partition the real line without truncation. At zero SE, deterministic threshold ties are a software convention, not a unit-mass Gaussian limit. The result bounds region disagreement, not student gain.

The intermediate certificate depends on both tails. For fixed SE, the Gaussian interval mass is maximized at the region midpoint. Thus no HAKD certificate exists when $\hat{\sigma}_H > (\tau_{\text{high}} - \tau_{\text{low}})/(2z_{1-\rho/2})$. At the default

TABLE XII
VCTK SEED-WISE SPEAKER PARTITIONS: CLASS COUNTS.

Split	English	Irish	Scottish	American	Total
Private, seed 0	90	90	60	120	360
Auxiliary, seed 0	30	30	60	0	120
Private, seed 1	60	90	120	90	360
Auxiliary, seed 1	60	30	0	30	120
Private, seed 2	90	120	60	90	360
Auxiliary, seed 2	30	0	60	30	120
Development/test	30	30	30	30	120

width 0.15 and $\rho = 0.10$, this SE limit is approximately 0.0456. Smaller SE permits certification, but the observed health must still be sufficiently far from each boundary. These are properties of the working Gaussian, not calibration guarantees for future speakers.

II. VCTK PROTOCOL AND RELEASE DIAGNOSTICS

A. Manifest and Privacy Accounting

The VCTK 0.92 subset [4] has 720 mono 48-kHz FLAC utterances, resampled to 16 kHz. Each seed repartitions 480 training utterances by speaker using Group-ShuffleSplit, auxiliary fraction 1/4; development/test stay fixed. All four sets are speaker/utterance-disjoint within seed. Table XII shows that seeds 0–2 each lack a different auxiliary class. The BS mel-CNN teacher uses four epochs, batch size 64, $C = 8$, $\sigma = 1$, and 24 private updates; Opacus 1.6 PRV gives $\varepsilon = 6.0829038771$ at $\delta = 10^{-5}$. Students train five auxiliary epochs. PyTorch 2.11 runs on an RTX 5090 with 32-bin, 80-frame mel features. Both models use AdamW (lr = 10^{-3} , wd = 10^{-4}); student $\alpha = 0.7$, $T = 2$.

Sampling-rate metadata: The six-batch loader gives Poisson $q = 1/\lceil 360/64 \rceil = 1/6$ and expected batch size 60. Cached histories record this rate and 24 updates; PRV replay matches within 10^{-9} . Accounting assumes add/remove-one-record adjacency and permitted fixed class-count metadata; student post-processing adds no budget.

B. Paired Modes and HAKD Weight Variants

Three seeds share teachers/caches and pair student initialization and auxiliary ordering. Main Table VI reports Hard, KD, instance HAKD, CTKD and S-Bacc; Table XIII adds the remaining variants. TCRD reuses Hard, and uniform HAKD equals KD. Balanced test classes make accuracy equal BalAcc.

Default HAKD uses $w(x) = \sum_k p_t^T(k | x) \hat{h}_k$, $T = 2$, development recalls (power 1, floor 0), and loss $(1 - \alpha w(x))\text{CE} + \alpha w(x)T^2\text{KL}(p_t^T \| p_s^T)$. Uniform/confidence/class variants use $w = 1$, $\max_k p_t^T(k | x)$, and $\hat{h}_y$. The difficulty probe weights CE by $a_i =$

TABLE XIII
ADDITIONAL VCTK STUDENTS (THREE-SEED MEAN $\pm$ SD).

Mode	BalAcc	Macro-F1	MajPred
CE-weighting probe	.214 $\pm$.043	.115 $\pm$.018	.786 $\pm$.185
Linear-T probe	.272 $\pm$.054	.153 $\pm$.070	.903 $\pm$.100
HAKD, confidence	.253 $\pm$.042	.185 $\pm$.057	.686 $\pm$.175
HAKD, class	.244 $\pm$.070	.178 $\pm$.084	.697 $\pm$.185
S-MF1	.239 $\pm$.021	.160 $\pm$.043	.733 $\pm$.221
S-MajPred	.253 $\pm$.043	.169 $\pm$.043	.761 $\pm$.128

TABLE XIV
VCTK SEED-WISE PAIRED TEST BALACC.

Seed	Hard	KD	HAKD	TCRD
0	0.3000	0.2500	0.2583	0.3000
1	0.2417	0.2500	0.2417	0.2417
2	0.2250	0.1833	0.2167	0.2250

$1 + (1 - \max_k p_t^T(k | x_i))$ and divides its weighted mean by $\bar{a}$, leaving KD unchanged. The linear-temperature probe uses $T_e = 2(1 + e/5)$, $e = 1, \dots, 5$. These probes are not reproductions of the cited adaptive methods.

S-Bacc/S-MF1 maximize student-dev BalAcc/Macro-F1; S-MajPred minimizes MajPred. Hard/KD/HAKD ties follow that order. Seed-wise choices are Hard/Hard/HAKD, HAKD/Hard/HAKD, and Hard/HAKD/HAKD, respectively. All teacher health values are below 0.45, yet KD gain is positive for seed 1.

Published CTKD mechanism: Global-T CTKD [5] learns $T = 1 + 20 \text{sigmoid}(u)$, initialized at $u = 1$. At epoch e , u 's backward multiplier is $-\gamma_e$, with $\gamma_e = [1 - \cos(\pi \min(e, 10)/10)]/2$; gradients include T^2 . CE/KL weights are 0.3/0.7. Student and temperature share AdamW ($10^{-3}, 10^{-4}$) for learning rate/weight decay, replacing the image optimizer while preserving its mechanism and ten-epoch ramp. VCTK uses five epochs; the small Common Voice archive uses 15, CV17 80 (with its main V optimizer). Temperature, loss, and gradients match the official implementation in float64.¹

C. Development Size and Threshold Sensitivity

Defaults are $\lambda = (1/3, 1/3, 1/3)$, thresholds (0.45, 0.60), and $\rho = 0.10$. Three class-stratified development sizes cross four prespecified threshold pairs. At each size/seed, retraining uses only that development subset for checkpoint/health decisions: nine teachers and 27 Hard/KD/HAKD students. Routes reuse paired branches; test utility is report-only. C counts certificates among three seeds; uncertified routes use Hard. Default policies agree; near-boundary fallback can improve or reduce utility. Earlier nominal-95% held-out-health

¹<https://github.com/zhengli97/CTKD>, commit 5611289.

TABLE XV
DEVELOPMENT-BUDGET AND THRESHOLD GRID: MEAN BALACC.

n_{dev}	Thresholds	TCRD	RC	C
30	(0.45,0.60)	0.2444	0.2444	3
60	(0.45,0.60)	0.2444	0.2444	3
120	(0.45,0.60)	0.2556	0.2556	3
30	(0.15,0.25)	0.2861	0.2444	1
60	(0.15,0.25)	0.2583	0.2444	1
120	(0.15,0.25)	0.2250	0.2389	1
30	(0.25,0.35)	0.2833	0.2444	0
60	(0.25,0.35)	0.2639	0.2444	0
120	(0.25,0.35)	0.2389	0.2417	1
30	(0.30,0.40)	0.2417	0.2444	1
60	(0.30,0.40)	0.2444	0.2444	0
120	(0.30,0.40)	0.2417	0.2556	1

TABLE XVI
VCTK COMPONENT SENSITIVITY (THREE STUDENTS; H: HARD, A: HAKD).

Weighting	$\hat{H} \pm \text{SD}$	Modes	Student BalAcc
Equal	0.271 ± 0.032	H(3)	0.256 ± 0.039
B only	0.222 ± 0.027	H(3)	0.256 ± 0.039
F only	0.164 ± 0.004	H(3)	0.256 ± 0.039
$1 - M$ only	0.428 ± 0.113	H(2), A(1)	0.242 ± 0.017
Leave B out	0.296 ± 0.058	H(3)	0.256 ± 0.039
Leave F out	0.325 ± 0.046	H(3)	0.256 ± 0.039
Leave $1 - M$ out	0.193 ± 0.011	H(3)	0.256 ± 0.039
Biased toward B	0.252 ± 0.014	H(3)	0.256 ± 0.039
Biased toward F	0.229 ± 0.021	H(3)	0.256 ± 0.039
Biased toward $1 - M$	0.334 ± 0.064	H(3)	0.256 ± 0.039

comparisons gave 2/3 and 3/5 coverage in overlapping snapshots; these are neither pooled nor treated as distribution-free calibration.

D. Health Components and Relation to Student Gain

Table XVI recomputes health on three teachers but uses independently trained selected-mode students, not Table XIV’s strict checkpoints. Single-component/leave-one-out weights use unit/equally split mass; biased settings permute (0.6, 0.2, 0.2) in ($B, F, 1 - M$) order. Most settings select the same route.

Recomputed strict-pair health-gain diagnostic: Table XVII uses exactly Table XIV’s teachers and students, with paired gains $-0.050000, +0.008333, -0.041667$. Three seeds cannot establish predictive increment; a composite-plus-components regression is rank-deficient, and a component-only fit is underdetermined.

E. Weak Mel-CE Teachers and a Heterogeneous Pair

VCTK Mel-CE seeds 0–4 at $\sigma = 1$ span health 0.1278–0.3123. Hard/KD/HAKD BalAcc is $0.2633 \pm 0.0451/0.2733 \pm 0.0484/0.2667 \pm 0.0598$. This fixed-backbone teacher-seed variation has positive mean KD gain, unlike the Common Voice grids.

TABLE XVII
VCTK HEALTH-GAIN CORRELATIONS (THREE PAIRED SEEDS).

Score	Pearson r	Spearman ρ
B	0.479	0.500
F	−0.609	−0.500
$1 - M$	−0.916	−1.000
H	−0.965	−1.000

TABLE XVIII
VCTK HUBERT-TO-MEL-CNN TRANSFER (THREE SEEDS).

Mode	BalAcc	Macro-F1
Hard	0.2306 ± 0.0411	0.1021 ± 0.0137
KD	0.2750 ± 0.0363	0.1408 ± 0.0485
CTKD, global	0.3111 ± 0.0625	0.1831 ± 0.0754
HAKD	0.2278 ± 0.0394	0.1055 ± 0.0144
TCRD (Hard)	0.2306 ± 0.0411	0.1021 ± 0.0137
S-BAcc	0.2722 ± 0.0394	0.1424 ± 0.0468

A frozen HuBERT-base encoder [6] (768 hidden units; 94,371,712 parameters) feeds a DP linear BS head and supervises a mel-CNN student. All three caches give $q = 1/6$, 24 updates, $\sigma = 1$, $C = 8$, $\delta = 10^{-5}$, and replay-verified PRV $\varepsilon = 6.0829038771$. Table XVIII tests frozen features, not end-to-end private representation learning. Teacher-only routing forgoes beneficial KD across architectures; student-dev selection retains most of its gain.

F. Private-Release Comparators and Runtime

Main Table VI compares direct release and matched PATE. A fixed ID hash assigns records to three disjoint shards, stable under insertion/deletion; non-private CE mel teachers train four epochs. For 120 auxiliary count-vector queries of sensitivity $\sqrt{2}$, Gaussian noise $\sigma = 13.683077843$ gives RDP $\varepsilon = 6.0829038771$, $\delta = 10^{-5}$, matching the single-teacher PRV bound. Features, students, initialization/order, five epochs, and dev selection match VCTK. PATE uses noisy-label CE; PATE + labels weights noisy/permitted-label CE 0.7/0.3. No raw ensemble weights/votes are released.

Matched full-training timings use small Common Voice seeds 0–2, $\sigma = 1$, resident features, fixed teachers, 15 student epochs, batch size 128, paired initialization/order. Each mode has a full warmup and three randomly ordered repeats, CUDA-synchronized at stage boundaries. Hard/KD/HAKD/TCRD/RC-TCRD/S-BAcc seconds (mean $\pm$ timing SD) are $.214 \pm .005/.276 \pm .007/.303 \pm .008/.218 \pm .005/.237 \pm .006/.790 \pm .017$ for SSL and $.727 \pm .017/.749 \pm .020/.834 \pm .024/.738 \pm .018/.757 \pm .018/2.319 \pm .056$ for Mel. SDs describe nine timings, not independent datasets. Health costs 5.9/12.4 ms (SSL/Mel), or 24.5/31.0 ms with RC resampling. Mean CUDA peaks, including teacher/student,

TABLE XIX
SPEAKER-DISJOINT IEMOCAP STUDENTS (FIVE SEEDS).

Mode	BalAcc	Macro-F1	MajPred
Hard	0.478 ± 0.007	0.446 ± 0.010	0.376 ± 0.022
KD	0.506 ± 0.014	0.496 ± 0.013	0.354 ± 0.016
HAKD	0.489 ± 0.008	0.471 ± 0.004	0.361 ± 0.041

are 18.1 MiB (SSL) and 389.2–389.4 MiB (Mel). S-BAcc trains/validates three candidates. Common teacher training and feature extraction are excluded.

III. ADDITIONAL PAIRED SPEECH EXPERIMENTS

A. IEMOCAP Feature Data and Record Unit

Four-category WavLM-large features come from D. Yohanes and A. Chowanda’s CC BY 4.0 release.² Removing 4,424 unidentified rows leaves 13,272 training rows; three variants of 4,424 utterances. With 553 validation and 554 test IDs, there are 5,531 utterances. Length-aware mean pooling followed by within-ID variant averaging yields one 1,024-dimensional record per ID; missing identities are never inferred.

Speaker identity uses the session plus utterance suffix: Ses01F_impro01_M000 belongs to Ses01M. Sessions 1–3/4/5-female/5-male supply private/auxiliary/development/test sets of 3,259/1,031/590/651 utterances, with no shared speakers or IDs. Released category IDs 0–3 are retained; the protected record is one utterance, not one variant.

B. Five-Seed Feature Routing Results

Linear BS teachers use auxiliary counts (327, 303, 258, 143), four epochs, batch size 64, $C = 1$, $\sigma = 1$, and Poisson $q = 1/51$: 204 updates give PRV $\epsilon = 1.8405159871$, $\delta = 10^{-5}$. Linear students use five epochs, $\alpha = 0.5$, $T = 2$. All heads use AdamW ($lr = 10^{-3}$, $wd = 10^{-4}$) and maximum development Macro-F1 checkpoints. Student initialization and ordering are paired. Health 0.3306–0.4238 yields Hard under both routes; all five RC decisions satisfy the model-region criterion. Both routes exclude useful KD. Five seeds measure optimization variation on this partition, not robustness across test speakers.

C. Common Voice Protocol and Additional Metrics

For main IV-A’s small archive (not CV17), seeds 0–4 have private sizes 4,340/4,529/4,525/5,576/5,215; remaining training clips are auxiliary. Poisson rates are

TABLE XX
SMALL COMMON VOICE MACRO-F1 (FIVE-SEED MEAN ± SD).

Teacher	σ	Hard	KD	HAKD	CTKD
SSL-BS	1.0	.303 ± .043	.273 ± .082	.325 ± .032	.285 ± .076
SSL-BS	1.5	.303 ± .043	.292 ± .061	.318 ± .043	.299 ± .040
SSL-BS	2.0	.303 ± .043	.282 ± .059	.318 ± .046	.280 ± .049
Mel-CE	1.0	.321 ± .039	.254 ± .009	.264 ± .024	.251 ± .006
Mel-CE	1.5	.321 ± .039	.255 ± .010	.251 ± .006	.251 ± .004
Mel-CE	2.0	.321 ± .039	.250 ± .006	.250 ± .004	.251 ± .006

1/34, 1/36, 1/36, 1/44, 1/41; update counts are eight times these denominators. PRV replay error is below 3×10^{-9} . BS uses auxiliary counts clamped to one.

Health 0.1469–0.3136 puts all 30 teachers below 0.45; RC certifies Hard throughout. At $\sigma = 1/1.5/2$, SSL S-BAcc/S-MF1/S-MajPred mean BalAcc triples are (.3515, .3490, .3389), (.3513, .3489, .3449), and (.3542, .3509, .3436); Mel triples are (.3529, .3520, .3520), (.3510, .3520, .3520), and (.3500, .3520, .3520). All selectors use development data only.

The additional Mel-BS comparison uses ten separate cached teachers, five seeds each at $\sigma = 1, 1.5$, with fixed permitted class-count metadata and the same 15-epoch student settings. Paired KD-minus-Hard differences are -0.0265 ± 0.0237 and -0.0262 ± 0.0145 , all ten negative. CTKD obtains $0.3394 \pm 0.0188/0.3326 \pm 0.0016$ balanced accuracy, versus Hard/TCRD’s $0.3564 \pm 0.0185/0.3556 \pm 0.0175$. These teachers are not pooled with Mel-CE.

REFERENCES

- [1] C. Dwork and A. Roth, *The Algorithmic Foundations of Differential Privacy*. Now Publishers Inc., 2014.
- [2] J. Ren *et al.*, “Balanced meta-softmax for long-tailed visual recognition,” in *Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 4175–4186.
- [3] A. K. Menon, S. Jayasumana, A. S. Rawat, H. Jain, A. Veit, and S. Kumar, “Long-tail learning via logit adjustment,” in *Proc. Int. Conf. Learn. Represent.*, 2021.
- [4] J. Yamagishi, C. Veaux, and K. MacDonald, “CSTR VCTK corpus: English multi-speaker corpus for CSTR voice cloning toolkit (version 0.92),” 2019, doi: 10.7488/ds/2645.
- [5] Z. Li *et al.*, “Curriculum temperature for knowledge distillation,” in *Proc. AAAI Conf. Artif. Intell.*, vol. 37, no. 2, 2023, pp. 1504–1512, doi: 10.1609/aaai.v37i2.25236.
- [6] W.-N. Hsu, B. Bolte, Y.-H. H. Tsai, K. Lakhotia, R. Salakhutdinov, and A. Mohamed, “HuBERT: Self-supervised speech representation learning by masked prediction of hidden units,” *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 29, pp. 3451–3460, 2021.
- [7] C. Busso *et al.*, “IEMOCAP: Interactive emotional dyadic motion capture database,” *Lang. Resour. Eval.*, vol. 42, no. 4, pp. 335–359, 2008, doi: 10.1007/s10579-008-9076-6.

²<https://doi.org/10.5281/zenodo.17803295>; raw USC IEMOCAP [7] has separate access terms.